

Understanding and Identifying Air Conditioner Usage from Electrical Current Data

To understand how electricity is used in a room, I analyzed a dataset containing one day of electrical measurements. The dataset includes 5575 readings with voltage, current, and timestamps. The current column represents the total electrical consumption of the room, meaning it combines the usage of all appliances such as lights, fans, and possibly an air conditioner.

Using basic electrical concepts, current indicates the flow of electricity used by appliances, while voltage represents the electrical supply. Together they help estimate the power consumption of devices. When exploring the dataset statistically, it became clear that most readings are very small. In fact, about 75% of the current values are below 0.36 A, which likely represents background electricity usage from low-power appliances like lights and fans.

However, the dataset also contains occasional spikes where the current increases significantly, reaching up to 25 A. These sudden increases indicate the presence of high-power appliances. Air conditioners typically consume much more electricity than small devices and operate in cycles where the compressor turns on, runs steadily for some time, and then turns off. This creates a recognizable pattern of a sudden increase in current followed by a sustained high level.

To isolate AC usage, a simple statistical method can be applied. The dataset shows an average current of 1.78 A with a standard deviation of 3.80 A. Based on this, a threshold around 5 A can be used to detect high-power events.

Conclusion

When the current exceeds this threshold and remains high for a longer duration, it likely indicates the operation of an air conditioner. By combining current magnitude and duration patterns, AC usage can be logically separated from the total electrical signal. This approach provides a practical way to prepare the data for further analysis or future energy monitoring systems.